# 1. Stepwise acid-base analysis

### Steps

- pH → primary process (HCO3 / PaCO2)
- Compensation: **Winter: expected PaCO2 = 1.5 × HCO3 + 8 (± 2) mmHg**
- Anion gap = Na − (Cl + HCO3), normal 8–12; **correct for albumin**: +2.5 per 10 g/L below 40
- Delta ratio (ΔAG/ΔHCO3): < 0.4 NAGMA, 1–2 pure HAGMA, > 2 + metabolic alkalosis
- Osmolar gap = measured − (2Na + glucose + urea), > 10 abnormal

**Key point:** Always correct the anion gap for albumin in ICU patients.

# 2. HAGMA differential (GOLDMARK)

## GOLDMARK

- **G**lycols (ethylene, propylene)
- **O**xoproline (pyroglutamic acid)
- **L**-lactate
- **D**-lactate (short bowel)
- **M**ethanol
- **A**spirin
- **R**enal failure
- **K**etoacidosis (diabetic, alcoholic, starvation)

## In pancreatitis + alcohol

- Lactate (shock, ischaemia), AKA, AKI, toxic alcohol co-ingestion, thiamine deficiency

**Key point:** Use GOLDMARK and then point to the likely ones in the stem.

# 3. Type A vs Type B lactic acidosis

### Type A (hypoperfusion/hypoxia)

- Shock, regional ischaemia (mesenteric), severe hypoxaemia, anaemia, CO poisoning

### Type B (no hypoperfusion)

- B1 disease: liver failure, sepsis (aerobic glycolysis, β2 stimulation), malignancy, thiamine deficiency
- B2 drugs: metformin, adrenaline, salbutamol, propofol infusion syndrome, linezolid, NRTIs, cyanide (nitroprusside)
- B3 inborn errors

### How to separate

- Perfusion markers (CRT, ScvO2, CO2 gap > 6 mmHg), lactate/pyruvate, response to resuscitation, drug history

**Key point:** A lactate that does not fall with resuscitation: think Type B or hidden ischaemia.

# 4. Management of severe metabolic acidosis

## Steps

- Treat the cause: resuscitate, source control, stop offending drugs, thiamine
- Ventilation: match respiratory compensation if intubated (avoid ↓ minute volume)
- Vasopressors for shock (catecholamine response falls at pH < 7.2)

## Bicarbonate: controversy

- Raises CO2 (needs ventilation), ↓ ionised Ca, hypernatraemia, overload, intracellular acidosis
- **BICAR-ICU 2018**: no overall benefit; pH ≤ 7.20 + AKI (KDIGO 2–3) → ↓ mortality and RRT
- Indicated: NAGMA with bicarbonate loss, hyperkalaemia, TCA/salicylate toxicity

## RRT

- Refractory acidosis, AKI, toxins (dialysable)

**Key point:** Quote BICAR-ICU: bicarbonate helps in pH ≤ 7.2 with AKI.

# 5. Pyroglutamic acidosis and propylene glycol

## Pyroglutamic (5-oxoproline)

- Glutathione depletion: chronic paracetamol + sepsis, malnutrition, liver/renal disease, flucloxacillin, vigabatrin; women
- HAGMA without lactate/ketones/toxic alcohol; urine organic acids (5-oxoproline)
- Stop paracetamol and flucloxacillin, give N-acetylcysteine

## Propylene glycol

- Solvent in IV lorazepam, diazepam, phenobarbital, etomidate
- High doses → HAGMA, raised **osmolar gap**, lactate (metabolised to L-lactate), AKI
- Monitor osmolar gap (> 10–12) in prolonged high-dose lorazepam; stop drug, RRT if severe

**Key point:** Unexplained HAGMA in ICU: check osmolar gap and drug chart.

# 6. Toxic alcohols

## Features

- Early ↑ osmolar gap → later ↑ anion gap
- Ethylene glycol: oxalate crystals, AKI, hypocalcaemia
- Methanol: visual loss, basal ganglia necrosis

## Treatment

- **Fomepizole** (or ethanol) to block alcohol dehydrogenase
- Haemodialysis (EXTRIP): severe acidosis, AKI, visual signs, high level
- Cofactors: thiamine + pyridoxine (EG), folinic acid (methanol); bicarbonate

**Key point:** High AG + high osmolar gap = toxic alcohol until proven otherwise.

# 7. AKI and RRT essentials

## KDIGO

- Stage 1: Cr × 1.5–1.9 or ↑ 26.5 µmol/L; UO < 0.5 mL/kg/h 6–12 h
- Stage 2: × 2–2.9; stage 3: × 3, Cr ≥ 354, RRT, UO < 0.3 × 24 h or anuria 12 h

## RRT timing and dose

- STARRT-AKI, AKIKI: early start no benefit; start for life-threatening indications (AEIOU)
- Dose 20–25 mL/kg/h effluent (RENAL, ATN)
- Citrate anticoagulation preferred for CRRT

**Key point:** No benefit from accelerated RRT: wait for a clear indication.